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Synthetic Data Augmentation to Improve Flow Zone Predictability for AI Application

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Flow Zone Indicator (FZI) is a derived petrophysical parameter critical for permeability grouping and hydraulic flow unit (HFU) characterization. It aids geoscientists and reservoir engineers in delineating reservoir heterogeneity, optimizing facies modeling, and ultimately informing field development strategies [Tiab and Donaldson, 2012]. Traditional FZI calculation relies on core-derived porosity and permeability, often calibrated against well logs. However, core data is expensive and sparse, especially in brownfields, exploration wells, or tight carbonate settings.

Machine learning (ML) techniques have shown significant promise in automating FZI estimation using well log inputs. Yet, ML's performance is closely tied to the availability of labeled training data. When data volume is limited or classes are imbalanced as commonly seen in reservoirs with uneven sampling across flow units ML models tend to overfit, yield unstable predictions, or fail to generalize [Zhou, 2012].

To address these challenges, we propose a robust workflow leveraging **bootstrapped synthetic data augmentation**. Bootstrapping, a statistical resampling technique, allows the generation of multiple training datasets by sampling with replacement. When applied carefully, it can increase the effective training sample size, capture statistical variance, and support uncertainty quantification without introducing artificial bias, provided that the underlying data distribution is respected.

Introduction

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This paper presents a hybrid workflow rooted in domain-driven petrophysical understanding and modern ML practices, demonstrating how synthetic data generation can unlock the full potential of data-constrained subsurface ML applications.

Methods

Dataset Overview

The study utilized well log datasets from a Middle Eastern carbonate reservoir. Five vertical wells with conventional logs (gamma ray, resistivity, neutron porosity, bulk density, sonic) and corresponding FZI labels derived from core analyses formed the core dataset. After preprocessing, 1,024 depth intervals were retained.

FZI was computed using the following industry-standard equation:

$$FZI = k^{0.5} \phi^e$$

where:

- k is permeability in millidarcies,
- ϕ is porosity (fractional),
- e is a reservoir-specific exponent (typically ~ 0.5 – 1.0) accounting for tortuosity and lithofacies variability.

Data Preprocessing

- **Depth Alignment:** Logs were re-sampled to 0.5 ft resolution, ensuring consistency across inputs.
- **Missing Values:** Depth intervals with $>30\%$ null values were removed; remaining missing entries were interpolated using spline interpolation guided by stratigraphic markers.
- **Feature Selection:** Features were selected based on correlation with FZI, variance inflation factor (VIF), and domain relevance. Resistivity, density-neutron separation, and delta transit time emerged as high-importance predictors.

Bootstrapped Synthetic Data Generation

We generated 100 bootstrapped datasets by sampling with replacement from the original data. Each bootstrap sample retained the same size ($n = 1,024$), maintaining class balance as much as possible. Special care was taken to avoid over-representation of rare FZI classes by applying stratified sampling across FZI quantiles (Q1 to Q4).

The procedure ensured:

- **Preservation of statistical moments** (mean, variance) within input features.
- **Geological plausibility** by constraining samples within known stratigraphic units using zone tags.

Bootstrapping also allowed us to:

- Estimate prediction uncertainty.

- Reduce model variance via ensemble predictions.

Model Training and Evaluation

Three supervised ML models were tested:

- Random Forest Regressor (RF)
- Gradient Boosting Regressor (GBR)
- Support Vector Regression (SVR)

Each model was trained on every bootstrap dataset. For final predictions, outputs were averaged across all models and bootstraps, forming an ensemble estimator.

Performance metrics:

- Coefficient of Determination (R^2)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Prediction Interval Width (PIW)

Cross-validation was done using 10-fold stratified CV repeated 3 times, with holdout wells for blind validation.

Results

Baseline vs. Bootstrapped Performance

Model	R^2 (Original Data)	R^2 (Bootstrapped)	ΔR^2	RMSE	MAE
RF	0.71	0.83	+0.12	0.42	0.31
GBR	0.74	0.85	+0.11	0.39	0.30
SVR	0.67	0.78	+0.11	0.48	0.35

Bootstrapped models consistently outperformed their counterparts, showing marked improvement in both accuracy and generalization.

Uncertainty Estimation

Figure 1 (hypothetical): A fan plot of predicted FZI intervals across depth for one test well, with 90% confidence bands derived from bootstrapped predictions. The spread of predictions was notably reduced in better-logged zones (e.g., mid-reservoir), while higher uncertainty correlated with thinly sampled or lithological complex intervals.

Overfitting Control

Learning curves revealed reduced train-test gaps for bootstrapped models, especially for GBR. This confirmed improved generalization and mitigated variance issues prevalent in small datasets.

Discussion

Integration with Field Development

Improved FZI predictions impact:

- **Facies classification:** Enhanced identification of reservoir vs. non-reservoir zones.
- **Permeability mapping:** More accurate input to static models.
- **Zonation schemes:** Supporting vertical well targeting and hydraulic fracture stage optimization.

In a field development context, reliable FZI maps derived from bootstrapped models reduce dependency on additional coring campaigns, offering substantial cost and time savings.

Geological Validity of Synthetic Samples

While bootstrapping is data-driven, domain oversight is critical. We observed that constraining synthetic sampling within geological zones (e.g., Units A, B, C) preserved sedimentological consistency. Blind tests without zone constraints led to implausible predictions (e.g., high FZI in known tight layers).

Computational Trade-offs

Training 100 ensemble models increased CPU time by ~5x but was manageable with parallel processing. Bootstrapping is not as compute-intensive as GAN-based data generation, making it suitable for operational workflows.

Limitations and Future Work

- Bootstrapping cannot add new geological trends—only mimic statistical properties.
- Future work should integrate physics-guided ML or hybrid geostatistical-ML methods.
- Application to 3D models and integration with Geocellular grids (e.g., via property co-kriging) is ongoing.

Conclusion

Bootstrapped synthetic data generation offers a pragmatic and scientifically sound approach to enhancing ML performance in subsurface FZI prediction under data scarcity. By combining petrophysical domain logic with ensemble ML workflows, the method balances accuracy, uncertainty quantification, and computational feasibility. This approach is particularly valuable in mature fields or frontier exploration settings where data acquisition is constrained.